徐炀个人简历

2026.10

单位：南方科技大学，计算机科学与工程系　　邮箱: sustech@sustech.edu.cn
职称：副教授　　主页: clcs-sustech.github.io

研究方向

徐炀，南方科技大学副教授，博士生导师。美国宾夕法尼亚州立大学博士，清华大学硕士和学士。广东省珠江人才计划青年拔尖人才，深圳市孔雀计划人才（C 类）。

徐炀博士的研究方向集中在计算语言学、自然语言处理、心理语言学和认知科学的交叉领域。致力于探索人类自然语言现象与内在认知、心理活动，以及心智意识现象的关联。目前主持南方科技大学计算语言学与意识科学（CLCS）实验室，研究团队主要关注：

- 人类自然语言与大型语言模型（LLMs）生成的区别。
- 语言现象对人脑认知活动的反映。
- 心理学与认知科学对人工智能技术的增强。

工作经历

副教授　　2023 年 6 月 – 至今
南方科技大学（SUSTech），计算机科学与工程系

助理教授　　2018 年 8 月 – 2023 年 5 月
美国圣地亚哥州立大学（San Diego State University），计算机科学系

教育背景

博士　　2013 年 8 月 – 2018 年 5 月
美国宾夕法尼亚州立大学（Penn State），信息科学与技术学院

硕士　　2010 年 9 月 – 2013 年 7 月
清华大学，心理学系

工学学士　　2006 年 9 月 – 2010 年 7 月
清华大学，电子工程系

科研项目

《基于信息熵理论的大语言模型幻觉问题研究》
项目负责人，深圳市自然科学基金面上项目，2024-2027

《基于熵的大语言模型生成评估》（已结题）
项目负责人，南方科技大学-华为可信智能系统创新实验室校企横向项目，2025

PI, NSF-CRII-HCC (Award ID: 2105192), $166K, 2021 – 2024
Modeling Computer-Mediated Task-Oriented Dialogues with Multi-Modality Information Theoretic Approaches

Co-PI, NIH (subaward to SDSU HealthLink), $50K, 2021 – 2022
An NLP Approach to Analyze the Past and Present, and Inform the Future of Health Disparities Research

发表论文

1. Zhichen Liu, Tianle Lun, Zhibin Wen, Hao An, Yulin Ou, Jianhui Xu, Hao Zhang, Wenyi Fang, Yang Zheng, and **Yang Xu**. Fast and accurate probing of in-training LLMs' downstream performances. In *Advances in Neural Information Processing Systems (NeurIPS 2026)*, 2026. (CCF-A)

2. Yongyuan Li and **Yang Xu**. Uniform information density-based preference optimization. In *Conference on Language Modeling (COLM 2026)*, 2026

3. Hao An and **Yang Xu**. Teaching LLMs to abstain via fine-grained semantic confidence reward. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (Main Conference)*, 2026. (CCF-B)

4. Yu Wang, Leyi Lao, Langchu Huang, Gabriel Skantze, **Yang Xu**, and Hendrik Buschmeier. Investigating the representation of backchannels and fillers in fine-tuned language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5319–5348, San Diego, California, United States, July 2026. Association for Computational Linguistics. (Oral; CCF-A)

5. Zhichen Liu, Yongyuan Li, and **Yang Xu**. Think in sentences: Explicit sentence boundaries enhance language model's capabilities. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2275–2288, San Diego, California, United States, July 2026. Association for Computational Linguistics. (Oral; CCF-A)

6. Yulin Ou, Yu Wang, **Yang Xu**, and Hendrik Buschmeier. Identifying the periodicity of information in natural language. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1161–1175, San Diego, California, United States, July 2026. Association for Computational Linguistics. (CCF-A)

7. Zhongyu Yang, Dannong Xu, Yonghan Zhang, Kefan Chen, Xinyi Wang, **Yang Xu**, Wei Pang, and Yingfang Yuan. Do vision and text cues exhibit evidential coupling? UFO: A benchmark for compositional multimodal reasoning in unified models. In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*, 2026. (CCF-A)

8. Hao An, Yibin Lou, Jiayi Guo, and **Yang Xu**. Purging the gray zone: Latent-geometric denoising for precise knowledge boundary awareness. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 2562–2576, San Diego, California, United States, July 2026. Association for Computational Linguistics. (CCF-A)

9. Zhichen Liu, Kaitong Qin, Linhan He, and **Yang Xu**. WaveDetect: Robust framework for machine-generated text detection via wavelet transform. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 8712–8727, San Diego, California, United States, July 2026. Association for Computational Linguistics. (CCF-A)

10. Jianjie Zheng, Zhichen Liu, Zhanyu Shen, Jingxiang Qu, Guanhua Chen, Yile Wang, **Yang Xu**, Yang Liu, and Sijie Cheng. Evaluating memory capability in continuous lifelog scenario. In *Findings of the Association for Computational Linguistics: ACL 2026*, 2026. (CCF-A)

11. Xiaoshan He, Xiaoqun Liu, Haodong He, Yu Wang, and **Yang Xu**. Eye movement features can predict human preferences on machine-generated texts. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 470–486, San Diego, California, United States, July 2026. Association for Computational Linguistics

12. Zhichen Liu, Yongyuan Li, **Yang Xu**, Yu Wang, Yingfang Yuan, and Zuhao Yang. Evaluating text generation quality using spectral distances of surprisal. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025

13. Zhongyu Yang, Junhao Song, Zhang Luo, Zuhao Yang, **Yang Xu**, Jingfen Lan, Yonghan Zhang, Wei Pang, Siyang Song, and Yingfang Yuan. Rechar: Revitalising characters with structure preserved and user-specified aesthetic enhancements. In *Proceedings of the SIGGRAPH Asia 2025 Technical Communications*, SA Technical Communications '25. Association for Computing Machinery, 2025

14. Lam Nguyen and **Yang Xu**. Reasoning for translation: Comparative analysis of chain-of-thought and tree-of-thought prompting for LLM translation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 259–275, Vienna, Austria, July 2025. Association for Computational Linguistics

15. **Yang Xu**, Yu Wang, Hao An, Zhichen Liu, and Yongyuan Li. Detecting subtle differences between human and model languages using spectrum of relative likelihood. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 10108–10121, Miami, Florida, USA, November 2024. Association for Computational Linguistics. (CCF-B)

16. Yu Wang, **Yang Xu**, Gabriel Skantze, and Hendrik Buschmeier. How much does nonverbal communication conform to entropy rate constancy?: A case study on listener gaze in interaction. In *Findings of the Association for Computational Linguistics ACL 2024*, pages 3533–3545. Association for Computational Linguistics, 2024. (CCF-A)

17. Zuhao Yang, Yingfang Yuan, **Yang Xu**, Shuo Zhan, Huajun Bai, and Kefan Chen. Face: Evaluating natural language generation with fourier analysis of cross-entropy. *Advances in Neural Information Processing Systems*, 36, 2024. (CCF-A, co-first author)

18. **Yang Xu** and Yang Cheng. Spontaneous gestures encoded by hand positions improve language models: An information-theoretic motivated study. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 9409–9424, 2023. (CCF-A)

19. **Yang Xu**, Yang Cheng, and Riya Bhatia. Gestures are used rationally: Information theoretic evidence from neural sequential models. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 134–140, Gyeongju, Republic of Korea, 2022. International Committee on Computational Linguistics. (CCF-B)

20. Bruce I. Gaynes, Mark B. Shapiro, Jie Xu, Abel Saju Augustine, **Yang Xu**, and et.al. Hierarchical data visualization of experimental erythrocyte aggregation employing cross correlation and optical flow applications. *Microvascular Research*, 143:104386, 2022

21. **Yang Xu**. Global divergence and local convergence of utterance semantic representations in dialogue. In *Proceedings of the Society for Computation in Linguistics*, volume 4, pages 116–124. 2021

22. **Yang Xu** and Zheng-sheng Zhang. Historical changes in semantic weights of sub-word units. In Adam Jatowt, Nina Tahmasebi, Yang Xu, and Lars Borin, editors, *Computational Approaches to Semantic Change*, pages 169–187. Language Science Press, 2021

23. M Alex Kelly, **Yang Xu**, Jesús Calvillo, and David Reitter. Which sentence embeddings and which layers encode syntactic structure? In *Proceedings of the 42nd Annual Meeting of the Cognitive Science Society*, pages 2375–2381, 2020. (acceptance rate: 18.1%; CCF-B)

24. **Yang Xu**, Jiasheng Zhang, and David Reitter. Treat the word as a whole or look inside? subword embeddings model language change and typology. In *Proceedings of the 1st International Workshop on Computational Approaches to Historical Language Change*, pages 136–145, Florence, Italy, 2019. Association for Computational Linguistics

25. **Yang Xu**, Jeremy Cole, and David Reitter. Linguistic alignment is affected more by lexical surprisal rather than social power. *Proceedings of the Society for Computation in Linguistics*, 2(1):349–352, 2019

26. **Yang Xu**, Jeremy Cole, and David Reitter. Not that much power: Linguistic alignment is influenced more by low-level linguistic features rather than social power. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, volume 1, pages 601–610, 2018. (acceptance rate: 24.5%; CCF-A)

27. **Yang Xu** and David Reitter. Information density converges in dialogue: Towards an information-theoretic model. *Cognition*, 170:147–163, 2018. (Impact factor: 3.65; CCF-B)

28. **Yang Xu** and David Reitter. Spectral analysis of information density in dialogue predicts collaborative task performance. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 623–633, Vancouver, Canada, 2017. (acceptance rate: 25%; CCF-A)

29. **Yang Xu** and David Reitter. Entropy converges between dialogue participants: Explanations from an information-theoretic perspective. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 537–546, Berlin, Germany, August 2016. Association for Computational Linguistics. (acceptance rate: 28.0%; CCF-A)

30. **Yang Xu** and David Reitter. Convergence of syntactic complexity in conversation. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 443–448, Berlin, Germany, August 2016. Association for Computational Linguistics. (acceptance rate: 28.0%; CCF-A)

31. **Yang Xu** and David Reitter. An evaluation and comparison of linguistic alignment measures. In *Proceedings of Cognitive Modeling and Computational Linguistics (CMCL)*, pages 58–67, Denver, CO, 2015. Association for Computational Linguistics

32. Alexander G. Ororbia II, **Yang Xu**, Vito D'Orazio, and David Reitter. Error-correction and aggregation in crowd-sourcing of geopolitical incident information. In N. Agarwal et al., editor, *Social Computing, Behavioral Modeling and Prediction*, volume 9021 of *Lecture Notes in Computer Science*, pages 381–387.

Springer, 2015

33. David Reitter, **Yang Xu**, Patrick Craven, Anikó Sándor, R. Chris Garrett, E. Vince Cross, and Jerry L. Franke. Cognitive models predicting surprise in robot operators. In *Proc. International Conference on Cognitive Modeling*, pages 190–191, Groningen, Netherlands, 2015

34. **Yang Xu** and Hong Li. Influence of visibility range and degree of urgency on efficiency of evacuation: Mediating effect of herding behavior. *Studies of Psychology and Behavior*, 13(3):311–319, 2015. (Published in Chinese)

35. Hong Li, **Yang Xu**, Shi Chen, and Anqi Gao. Effects of intuition and analysis on the adoption of sources of information in escape decision-making. *Studies of Psychology and Behavior*, 10(6):452–458, 2012. (Published in Chinese)

会议报告

1. Global Divergence and Local Convergence of Dialogue, 线上, *SCiL*, 2021
2. Subword Embeddings Reveal Language Change, 海报, *CogSci*, 加拿大蒙特利尔, 2019
3. Treat the Word As a Whole or Look Inside, 海报, *ACL*, 意大利佛罗伦萨, 2019

特邀报告

加拿大滑铁卢大学 2022 年 8 月
邀请人: Moojan Ghafurian 博士
题目: Speaking Rationally by Gestures: Evidence from Multi-Modal Language Models

北京语言大学 2021 年 8 月
题目: The Information-Theoretic Insights of Multimodal Language Models

西湖大学 2019 年 11 月
邀请方: 张岳博士课题组 题目: Dialogue from an Information-Theoretic Perspective

上海交通大学 2019 年 11 月
邀请方: 工学院 题目: Efficient Communication in Dialogue

清华大学 2019 年 11 月
邀请方: 心理学系 题目: An Information Theoretic Psycholinguistic Model of Dialogue

San Diego State University 2019 年 10 月
邀请方: SDSU 人工智能研讨会 题目: Recent Advances in Natural Language Processing

San Diego State University 2018 年 2 月
邀请方: 计算机系 题目: Computational Understanding of Dialogue and Language

荣誉与奖励

研究卓越奖 宾夕法尼亚州立大学信息科学与技术学院，2018
优秀毕业生奖 清华大学，2013
优秀硕士学位论文奖 清华大学，2013

学术服务

ACL 2025–2026 领域主席（Area Chair）

CogSci 2018–2020 审稿人

Lingua 审稿人，*Dialogue and Discourse* 审稿人，2022

教学

@SUSTech

CS310 Natural Language Processing，自然语言处理（本科生） 2024 (春季)–2026

CS5026 Introduction and Frontiers of Cognitive Sciences，认知科学导论与前沿（研究生） 2025 (秋季) – 2026

CS208 Algorithms，算法设计与实现（本科生） 2023

@SDSU

CS460 Algorithms 2018 – 2022

CS549 Machine Learning 2018 – 2022